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**B. Tech. Degree VI Semester Special Supplementary Examination
November 2014**

**ME 604 HEAT AND MASS TRANSFER
(2006 Scheme)**

Time: 3 Hours

Maximum Marks: 100

**PART A
(Answer ALL questions)**

(8 x 5 = 40)

- I. (a) Derive an expression for critical radius of a cylinder.
- (b) Explain lumped heat capacity system and significance of Biot number Bi and Fourier number Fo.
- (c) Differentiate between drop wise condensation and film wise condensation.
- (d) What is Reynolds analogy? Describe the relation between the fluid friction and heat transfer.
- (e) Write a note on radiation shape factor. State and explain reciprocity theorem.
- (f) State and explain Kirchhoff's law for heat radiation.
- (g) What are the different types of heat exchangers?
- (h) Write a note on NTU method. What are the advantages of NTU method over LMTD method in a heat exchanger design?



PART B

(4 x 15 = 60)

- II. (a) Derive the general heat conduction equation in Cartesian coordinate system. (10)
 - (b) Deduce the general heat conduction equation in Cartesian coordinate for one dimensional steady state heat conduction with heat generation and show that the temperature distribution is parabolic. (5)
- OR**
- III. (a) A cylindrical ingot 8 cm diameter and 25 cm long passes through a heat treatment furnace which is 5 m in length. The ingot must reach a temperature of 850°C before it comes out of the furnace. The furnace gas is at 1200°C and ingot initial temperature is 100°C. What is the time taken by the ingot through furnace to attain the required temperature? The combined radiative and convective surface heat transfer coefficient is 100 W/m² °C. Take k (steel) = 40 W/m °C and (thermal diffusivity of steel) $\alpha = 1.16 \times 10^{-5}$ m² /s. Also find average velocity of ingot. (9)
 - (b) Write a note on transient heat conduction in semi infinite solids and the significance of Heisler chart and Gaussian error function in this analysis. (6)
- IV. A glass-door fire screen, used to reduce exfiltration of room air through a chimney has a height of 0.7 m and a width of 1 m and reaches a temperature of 227°C. If the room temperature is 28°C, estimate the free convection heat rate to the room. (15)

OR

(P.T.O.)

- V. Air at 25°C and at atmospheric pressure flows at a velocity of 2 m/s over a plate maintained at 95°C. The length and width of the plate are 800 mm and 400 mm respectively. Calculate the heat transfer rate from (i) first half of the plate (ii) next half of the plate. (15)

- VI. A paint baking oven consists of a long, triangular duct in which a heated surface is maintained at 1300K and another surface is insulated. Painted panels which are maintained at 500K occupy the third surface. The triangle is of width $W = 1$ m on a side and the heated and insulated surfaces have an emissivity of 0.7. The emissivity of the panel is 0.4. During steady state operation, at what rate must energy be supplied to the heated side per unit length of the duct to maintain its temperature at 1300K? What is the temperature of the insulated surface? (15)

OR

- VII. (a) Define the following terms related to thermal radiation. (i) Total emissive power (ii) Emissivity (iii) Irradiation (iv) Radiosity. (6)
(b) Show that the net radiation heat transfer is halved when inserting a single radiation shield between two parallel surfaces. Assume all the surfaces have the same emissivity. Use the electrical network analogy. (9)

- VIII. (a) Derive an expression for LMTD of a counter flow heat exchanger. (9)
(b) What are the fouling factors? How do they affect the heat exchanger design? Explain. (6)

OR

- IX. 16.5 kg/s of the product at 650°C ($C_p = 3.55$ kJ/kg°C), in a chemical plant, are to be used to heat 20.5 kg/s of the incoming fluid from 100°C ($C_p = 4.2$ kJ/kg°C). If the overall heat transfer coefficient is 0.95 kW / m² °C and the installed heat transfer surface is 44 m², calculate the fluid outlet temperatures for the counter flow and parallel flow arrangements. (15)
